

Foundations of Data Science

Lecture 2: Data and exploratory analysis

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Data

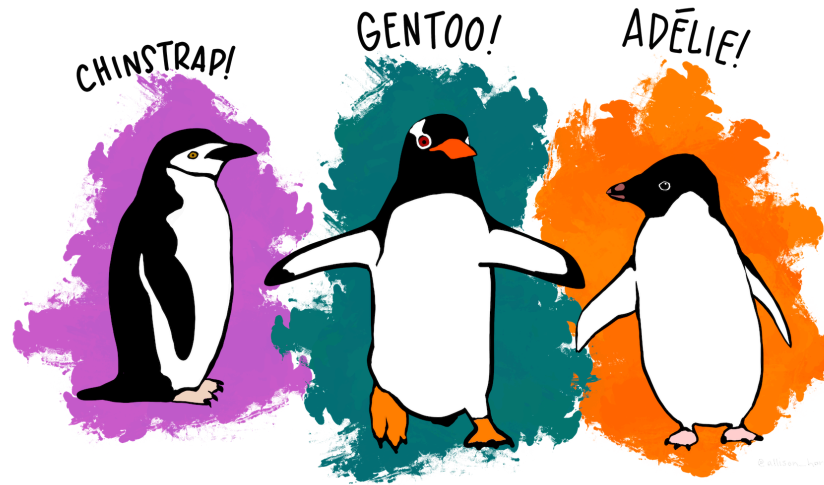


What is data?

Data are **recorded observations** about the world.

They can take many forms, including numbers, text, images, and more.

A running example



We will follow one dataset through this lecture: the **Palmer Archipelago penguins**, 344 birds of three species measured on three islands between 2007 and 2009.

Someone stood in the cold, caught a penguin, measured its bill with a caliper and put it on a scale. What follows is about what such records are, and what they are not.

Mathematically, data can be viewed as a function f that maps real-world entities ω to measurable values $\mathbf{x}$,

$$f : \Omega \rightarrow \mathcal{X},$$

where

- Ω is the sample space of possible states ω of the world,
- $\mathcal{X}$ is the measurement space of possible observations $\mathbf{x}$.

Examples:

- Penguin body mass: $\omega \in \{\text{penguins}\} \rightarrow x \in \mathbb{R}^+$ (g)
- Stock price: $\omega \in \{\text{market states}\} \rightarrow x \in \mathbb{R}^+$ (USD)
- Pixel colour: $\omega \in \{\text{scenes}\} \rightarrow \mathbf{x} \in \{0, \dots, 255\}^3$ (RGB values)

If the sample space Ω carries a probability distribution $p(\omega)$, then f turns a random state of the world into a random observation $\mathbf{x} = f(\omega)$. Its distribution is the one f induces from p , called the **data distribution** $p_r(\mathbf{x})$, where r stands for "real".

When $\mathcal{X}$ is continuous, it can be written as

$$p_r(\mathbf{x}) = \int_{\omega \in \Omega} p(\omega) \delta(\mathbf{x} - f(\omega)) d\omega,$$

where δ is the Dirac delta function. When $\mathcal{X}$ is discrete, p_r assigns probabilities rather than a density.

The **measurement process** is part of the data generation mechanism. We make it explicit by adding the measurement conditions $\xi \in \Xi$ (instrument settings, environmental conditions, observer effects) to the map,

$$f : \Omega \times \Xi \rightarrow \mathcal{X}.$$

Measurements can introduce quantization (continuous to discrete), noise (random perturbations), and bias (systematic deviations). If $\Omega \times \Xi$ carries a joint distribution $p(\omega, \xi)$, then

$$p_r(\mathbf{x}) = \iint_{\omega \in \Omega, \xi \in \Xi} p(\omega, \xi) \delta(\mathbf{x} - f(\omega, \xi)) d\omega d\xi,$$

which captures the variability of both the phenomenon and its measurement.



Example: how a penguin record is made.

- Bill length and depth: dial calipers, to 0.1 mm.
- Flipper length: ruler, to 1 mm.
- Body mass: Pesola spring scale and a weigh bag, to 25 g.
- Sex: not measured at all, but inferred in the lab from a blood sample.

Nests are found at the one-egg stage, both adults are caught, measured, sampled and released. All of this is ξ .

The instruments are visible in the data: every body mass is a multiple of 25 g, every bill length has one decimal, every flipper length is a whole millimetre.

Data types

Atomic data are the indivisible units of information collected through measurements. They are often categorized based on their nature and the operations that can be performed on it.

- Numerical (continuous, discrete)
- Categorical (nominal, ordinal)

Numerical data

- Continuous: $x \in \mathbb{R}$ (e.g., temperature), $x \in \mathbb{R}^+$ (e.g., height, weight)
- Discrete: $x \in \mathbb{N}$ (e.g., counts), $x \in \mathbb{Z}$ (e.g., differences)

Categorical data

- Nominal: $x \in \mathcal{C} = \{c_1, c_2, \dots, c_K\}$ (e.g., colors, types, text characters) without intrinsic order
- Ordinal: $x \in \mathcal{C}$ with ordering relations $c_1 \prec c_2 \prec \dots \prec c_K$ (e.g., ratings, grades)



Example: one penguin record

```
species: Adelie          # Categorical, nominal
island: Torgersen        # Categorical, nominal
bill length: 39.1 mm    # Numerical, continuous
body mass: 3750 g       # Numerical, continuous
sex: male                # Categorical, nominal
year: 2007               # Numerical, discrete
```

No variable here is ordinal; a rating from 1 to 5, or a grade, would be.

Data structures

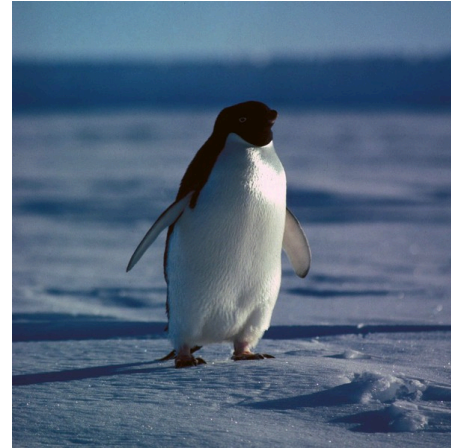
A measurement x can be a single atomic value or a composite structure made of multiple atomic values. Common aggregates or data structures include:

- Tabular data
- Arrays and tensors
- Sequences
- Networks and graphs

Tabular data

	species	island	bill length (mm)	body mass (g)	sex
0	Adelie	Torgersen	39.1	3750.0	male
1	Adelie	Torgersen	39.5	3800.0	female
2	Adelie	Torgersen	40.3	3250.0	female
3	Adelie	Torgersen	NaN	NaN	NaN
4	Adelie	Torgersen	36.7	3450.0	female
...	...	...	...	...	...
339	Chinstrap	Dream	55.8	4000.0	male
340	Chinstrap	Dream	43.5	3400.0	female
341	Chinstrap	Dream	49.6	3775.0	male
342	Chinstrap	Dream	50.8	4100.0	male
343	Chinstrap	Dream	50.2	3775.0	female

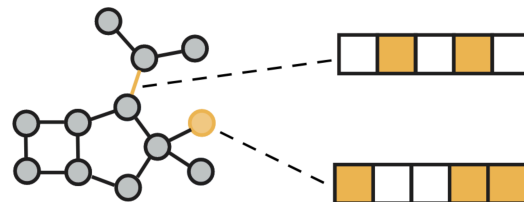
Arrays and tensors



Sequences

```
['P', 'e', 'n', 'g', 'u', 'i',  
 'n']
```

Networks and graphs



A data frame $\mathbf{X}$ represents a **tabular collection** of N records (rows) over d variables/atomic measurements (columns),

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{pmatrix}.$$

Each entry x_{ij} corresponds to the value of variable j for record i .

Variables are often heterogeneous (mixing numerical and categorical types). When all variables are numerical, the data frame can be viewed as a matrix $\mathbf{X} \in \mathbb{R}^{N \times d}$.

For the penguins, $N = 344$ records over $d = 8$ variables, four of them numerical.

Notation

Bold marks an object holding more than one number. Plain marks a single one.

- $\mathbf{X} \in \mathbb{R}^{N \times d}$ is the data frame, with N records indexed by i and d variables indexed by j .
- $\mathbf{x}_i$ is a record, the i -th row; $\mathbf{x}_j$ is the column of variable j ; the index letter says which.
- x_{ij} is a single entry, x_j a single variable seen as a random quantity.
- $\mathbf{x}$ is a generic measurement, $\mathcal{X}$ the space it lives in.

Collections of homogeneous measurements can be represented as **arrays** or **tensors** $\mathbf{X} \in \mathbb{R}^{d_1 \times d_2 \times \dots \times d_k}$, where the position of each atomic value in the array is usually associated to a spatial or temporal location.

- Images: 3d arrays $\mathbf{X} \in \{0, \dots, 255\}^{h \times w \times c}$ (height, width, channels). The photo on the previous slide is such an array.
- Videos: 4d arrays $\mathbf{X} \in [0, 255]^{t \times h \times w \times c}$ (time, height, width, channels).

Data can also be structured as ordered **sequences** $S = (x_1, x_2, \dots, x_T)$ indexed by time or position. Each element x_t can be atomic or composite.

- Time series: $S = (x_1, x_2, \dots, x_T)$ where x_t is a measurement at time t (e.g., stock prices).
- Text: $S = (w_1, w_2, \dots, w_T)$ where w_t is the t -th word in a document.

Finally, data can be organized as **networks** or **graphs** $G = (V, E)$, where entities are represented as nodes V and relationships as edges E . Each may also have associated attributes, x_v for nodes and x_{uv} for edges.

- Molecular structures, where nodes represent atoms and edges represent bonds.
- Social networks, where nodes represent individuals and edges represent interactions or relationships.



Follow the tutorials in `nb02a-tables.ipynb` , `nb02b-jax.ipynb` , and `nb02c-data-wrangling.ipynb` to practice working with arrays and data frames in Python.

Data quality

Real-world data are often imperfect and may suffer from various **quality issues** that can impact analysis and modeling. Common data quality issues include:

- Missing values
- Measurement errors
- Outliers

Missing values are common in real-world datasets and can arise from various factors such as non-response in surveys, sensor malfunctions, or data corruption.

Let $\mathbf{X}_{\text{full}}$ be the complete data and $\mathbf{M} \in \{0, 1\}^{N \times d}$ the missingness pattern, with $m_{ij} = 1$ when entry ij is observed and $m_{ij} = 0$ when it is missing¹.

What we actually hold is the pair $(\mathbf{X}_{\text{obs}}, \mathbf{M})$, where $\mathbf{X}_{\text{obs}} = \{x_{ij} : m_{ij} = 1\}$ are the observed entries and $\mathbf{X}_{\text{mis}} = \{x_{ij} : m_{ij} = 0\}$ the missing ones. A missing entry is not a zero, and no arithmetic can recover it.

1: Rubin's convention is the opposite, 1 for missing.

	species	island	bill len.	bill depth	flipper len.	body mass	sex
3	Adelie	Torgersen	NaN	NaN	NaN	NaN	NaN
8	Adelie	Torgersen	34.1	18.1	193.0	3475.0	NaN
9	Adelie	Torgersen	42.0	20.2	190.0	4250.0	NaN
10	Adelie	Torgersen	37.8	17.1	186.0	3300.0	NaN
11	Adelie	Torgersen	37.8	17.3	180.0	3700.0	NaN
47	Adelie	Dream	37.5	18.9	179.0	2975.0	NaN
178	Gentoo	Biscoe	44.5	14.3	216.0	4100.0	NaN
218	Gentoo	Biscoe	46.2	14.4	214.0	4650.0	NaN
256	Gentoo	Biscoe	47.3	13.8	216.0	4725.0	NaN
268	Gentoo	Biscoe	44.5	15.7	217.0	4875.0	NaN
271	Gentoo	Biscoe	NaN	NaN	NaN	NaN	NaN

The 11 penguin records with missing entries: two birds were never measured, nine have no recorded sex.

The pattern $\mathbf{M}$ is itself produced by the measurement process, and is modeled as such:

- Missing completely at random (MCAR): the pattern is independent of the data, $p(\mathbf{M} \mid \mathbf{X}_{\text{full}}) = p(\mathbf{M})$.
- Missing at random (MAR): the pattern may depend on what is observed, but not on what is missing, $p(\mathbf{M} \mid \mathbf{X}_{\text{full}}) = p(\mathbf{M} \mid \mathbf{X}_{\text{obs}})$.
- Missing not at random (MNAR): the pattern depends on the missing entries themselves.

Why the distinction matters: what we can compute is

$$p(\mathbf{X}_{\text{obs}}, \mathbf{M}) = \int p(\mathbf{X}_{\text{full}})p(\mathbf{M} \mid \mathbf{X}_{\text{full}})d\mathbf{X}_{\text{mis}}.$$

Under MCAR or MAR, the second factor does not depend on the missing entries and, provided it shares no parameters with the first, it can be ignored: modelling the observed data is enough. Under MNAR it cannot be ignored, and the mechanism must be modeled jointly with the data.

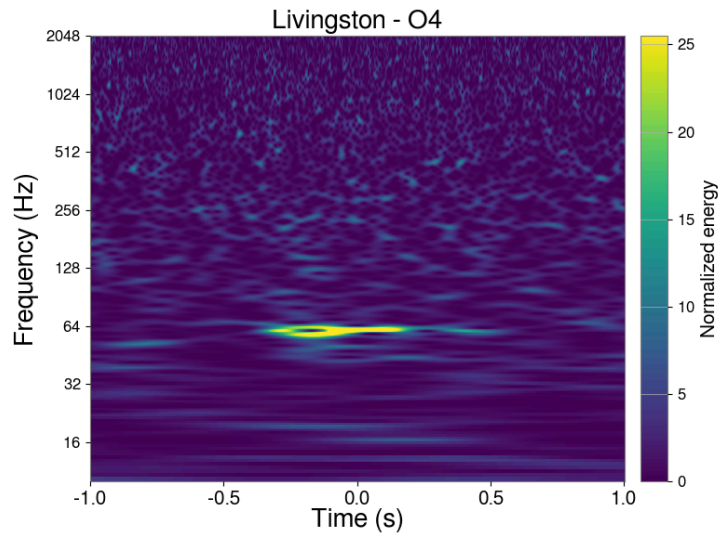
Example:

Survey of $n = 1000$ respondents, 30% do not answer the income question.

- MCAR: Randomly selected respondents skip the question.
- MAR: Younger respondents are less likely to answer.
- MNAR: High earners refuse to answer.

Imputing or discarding missing values without making explicit the assumptions about the missingness mechanism can lead to biased results.

The measurement process can also introduce acquisition errors and produce observations that deviate significantly from regular measurements. These **outliers** can arise from instrument malfunctions, data entry errors, or rare events.



Example:

Glitches in gravitational wave detectors are outliers that can mimic true signals and complicate detection efforts. They can arise from environmental disturbances, instrumental artifacts, or other non-astrophysical sources.

Treating outliers requires a model of the measurement process that either describes measurements under normal conditions or explicitly accounts for anomalies.

Two models make this explicit: a contamination mixture

$$p(\mathbf{x}) = (1 - \varepsilon)p_{\text{model}}(\mathbf{x}) + \varepsilon p_{\text{bad}}(\mathbf{x}),$$

which gives bad measurements their own distribution, or a heavy-tailed data model, which allows rare large deviations without special-casing them.

Outliers should not be removed blindly unless explicitly justified by the measurement model or domain knowledge.



Temperature readings in Liège:

20.1, 19.8, 20.3, 1000.0, 20.2, 19.9

The 1000.0 value is an outlier likely due to a sensor error.

19.2, 19.8, 20.3, 37.8, 20.2, 19.9

The 37.8 value is a rare but plausible measurement on a hot day.

Where this leaves us

Everything in this part is one picture of how the data came to be:

- the map f from states of the world to measurements,
- the conditions ξ under which measurements are taken,
- the pattern $\mathbf{M}$ of what ends up recorded,
- the data distribution $p_r(x)$ that all of this induces.

Exploratory data analysis

Exploratory data analysis (EDA) is the systematic examination of data to understand its structure, patterns and anomalies before formal modeling. It typically involves visual and quantitative techniques to summarize key characteristics of the data.

The goal is to generate hypotheses and inform modeling decisions, not to confirm preconceived notions.

What we are looking at

All we hold is N records, the rows $\mathbf{x}_i = (x_{i1}, \dots, x_{id})$ of the data frame. Their **empirical distribution** puts equal mass on each of them,

$$\hat{p}_N(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N \delta(\mathbf{x} - \mathbf{x}_i).$$

Every plot and every statistic that follows is a functional of $\hat{p}_N$: a histogram is a marginal, a scatter plot a joint, body mass by species a conditional, correlation and mutual information measure dependence, and PCA is a projection.

$\hat{p}_N$ is not p_r . It is what N records show of it, and that gap is what the rest of the course is about.



Back to the penguins, with the whole table in hand: 344 records, four numerical measurements, three species and three islands.

Switch to `nb02d-eda.ipynb` to follow along.

Univariate analysis

Let us consider a variable j of the data frame. We write x_j for that variable as a random quantity, distributed under p_r , and $\mathbf{x}_j = (x_{1j}, x_{2j}, \dots, x_{Nj})^T$ for the column of values actually recorded.

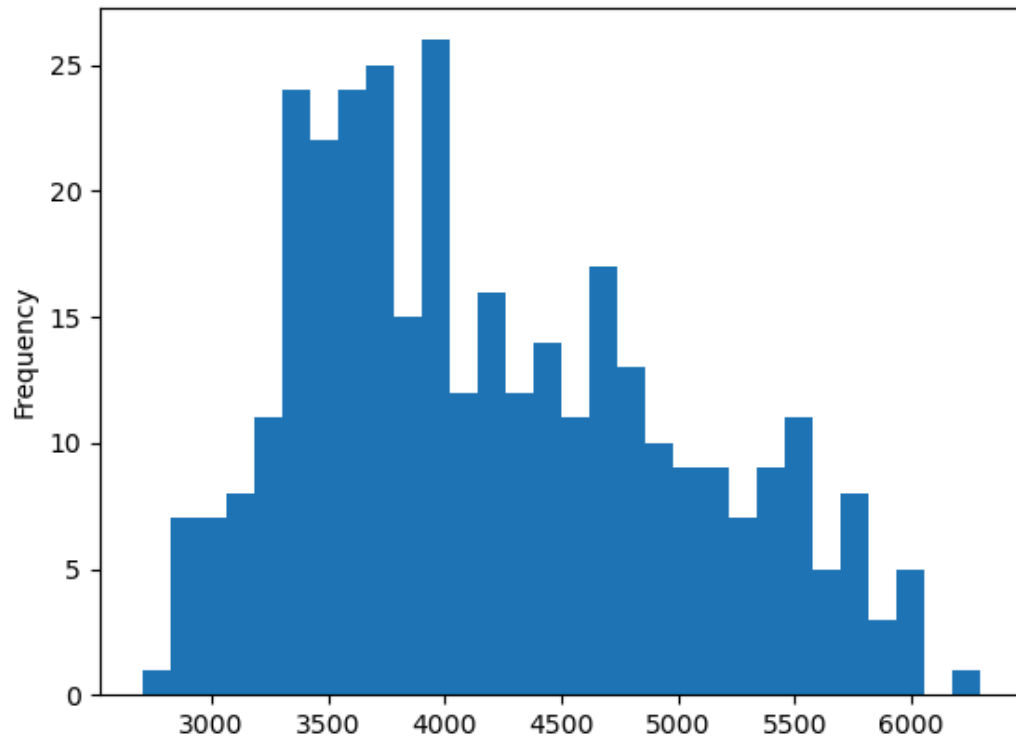
Univariate analysis focuses on understanding the distribution and characteristics of this single variable.

For numerical variables, common techniques include:

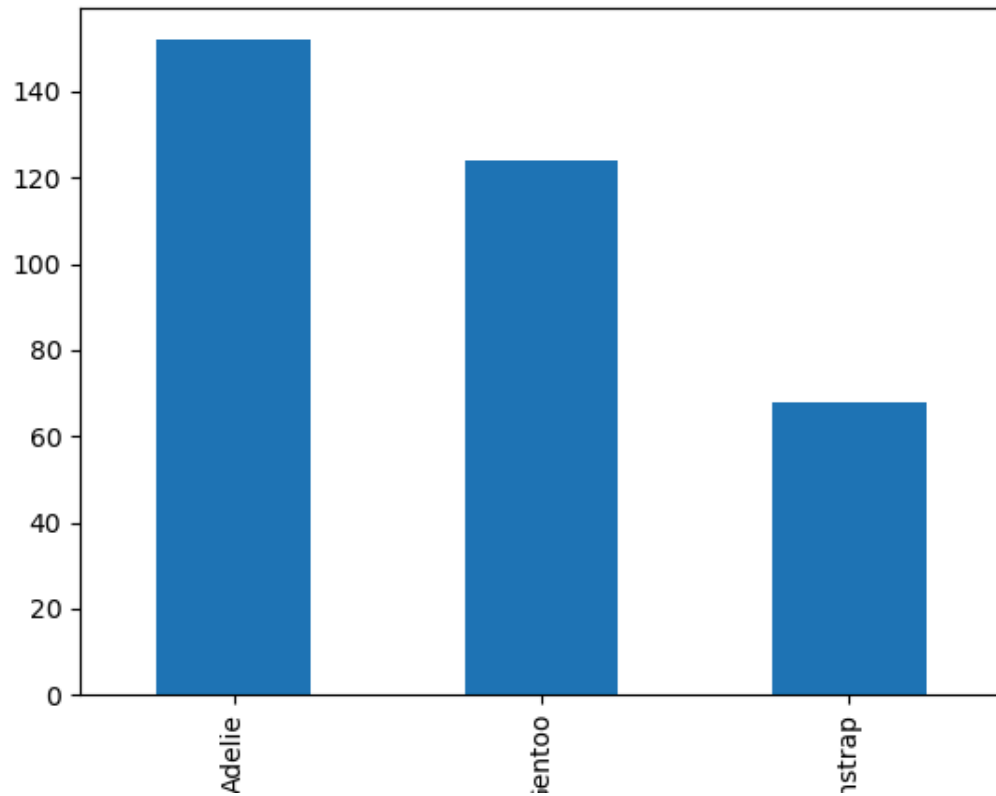
- Looking at the raw data: print values, scroll through them.
- Plotting the data: histograms reveal the distribution shape.
- Summarizing with statistics: mean, median, mode, variance, skewness, kurtosis.

For categorical variables:

- Counting occurrences of each category.



Numerical: Histogram of body mass for all penguins.
One broad mode with a long right tail: perhaps not a single population.



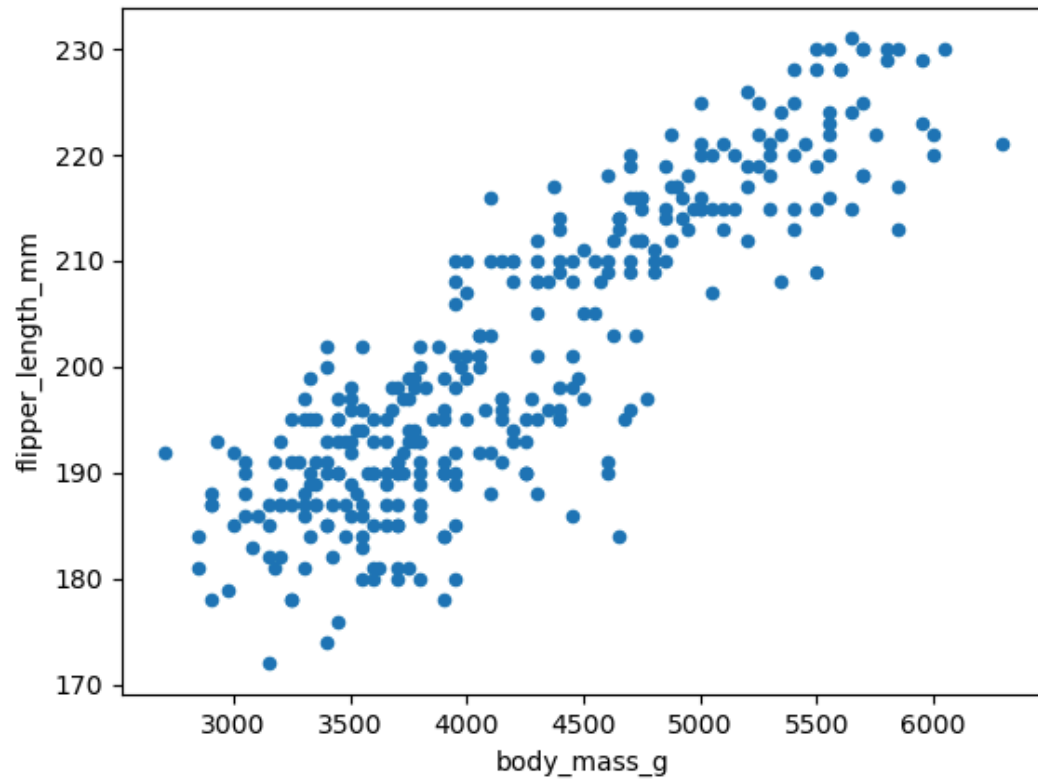
Categorical: Bar plot of species counts for all penguins.
Unbalanced groups: 152 Adelie, 124 Gentoo, 68 Chinstrap.

Bivariate analysis

Bivariate analysis examines the relationship between two variables j and k , from their recorded columns $\mathbf{x}_j$ and $\mathbf{x}_k$.

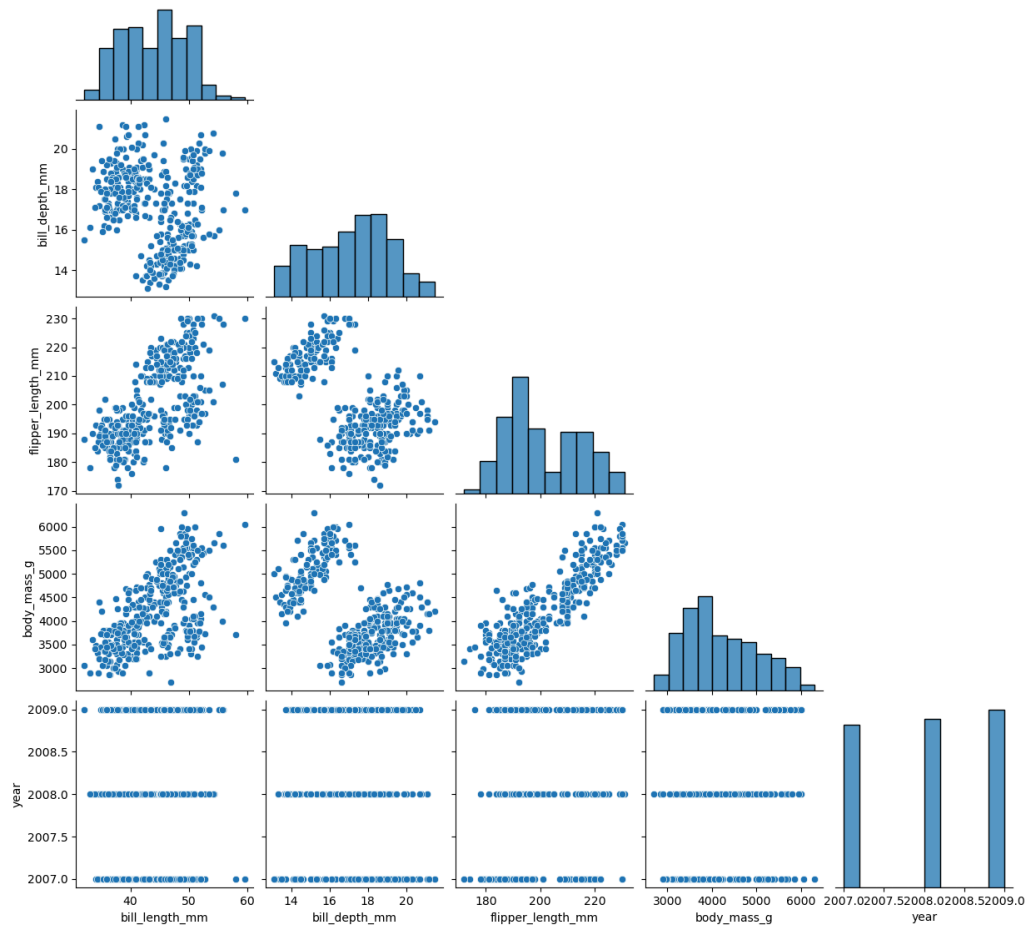
Depending on the types of variables, different techniques are used:

- Pair plots for two numerical variables (scatter or 2d histogram).
- Multiple histograms for categorical vs numerical variables.
- Contingency tables for two categorical variables.
- Correlation coefficients (e.g., Pearson, Spearman) for numerical variables.



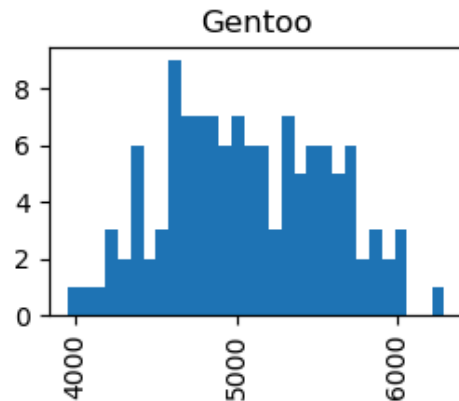
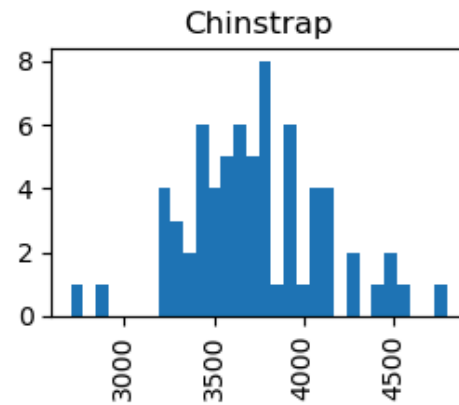
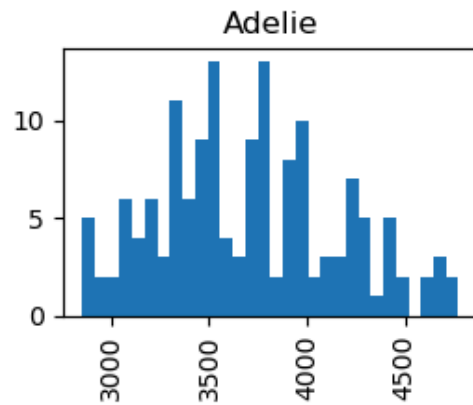
Numerical vs. numerical: Scatter plot of body mass vs flipper length.

Mass grows with flipper length, close to linearly.



Pair plots of all numerical variables.

Most panels show two or three clouds rather than one.



Categorical vs. numerical: Histograms of body mass by species.
The right tail was Gentoo, at 5076 g on average against about 3700 g for the others.

island	Biscoe	Dream	Torgersen
species			
Adelie	44	56	52
Chinstrap	0	68	0
Gentoo	124	0	0

Categorical vs. categorical: Contingency table of species and island.
Gentoo live only on Biscoe, Chinstrap only on Dream: species and island are dependent.

Correlation coefficients can quantify the dependency between two numerical variables. They are useful but come with assumptions and limitations.

- Pearson correlation measures linear relationships. Its empirical version reads

$$\hat{\rho}_{jk} = \frac{\sum_{i=1}^N (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)}{\sqrt{\sum_{i=1}^N (x_{ij} - \bar{x}_j)^2} \sqrt{\sum_{i=1}^N (x_{ik} - \bar{x}_k)^2}},$$

an estimate of $\rho_{jk} = \text{cov}(x_j, x_k) / (\sigma_j \sigma_k)$ under p_r . It ignores non-linear dependencies.

- Spearman correlation is the Pearson correlation of the rank-transformed variables. It captures monotonic relationships.
- Correlation does not imply causation and can be affected by outliers.

The **mutual information** between two variables x_j and x_k measures the reduction in uncertainty about one given knowledge of the other. For discrete variables,

$$I(x_j; x_k) = \sum_{x_j} \sum_{x_k} p_r(x_j, x_k) \log \frac{p_r(x_j, x_k)}{p_r(x_j)p_r(x_k)},$$

with integrals in place of the sums for continuous ones.

Mutual information captures any statistical relationship, not just linear or monotonic ones. It is a property of p_r , however, which must itself be estimated from the N records, and that is hard.

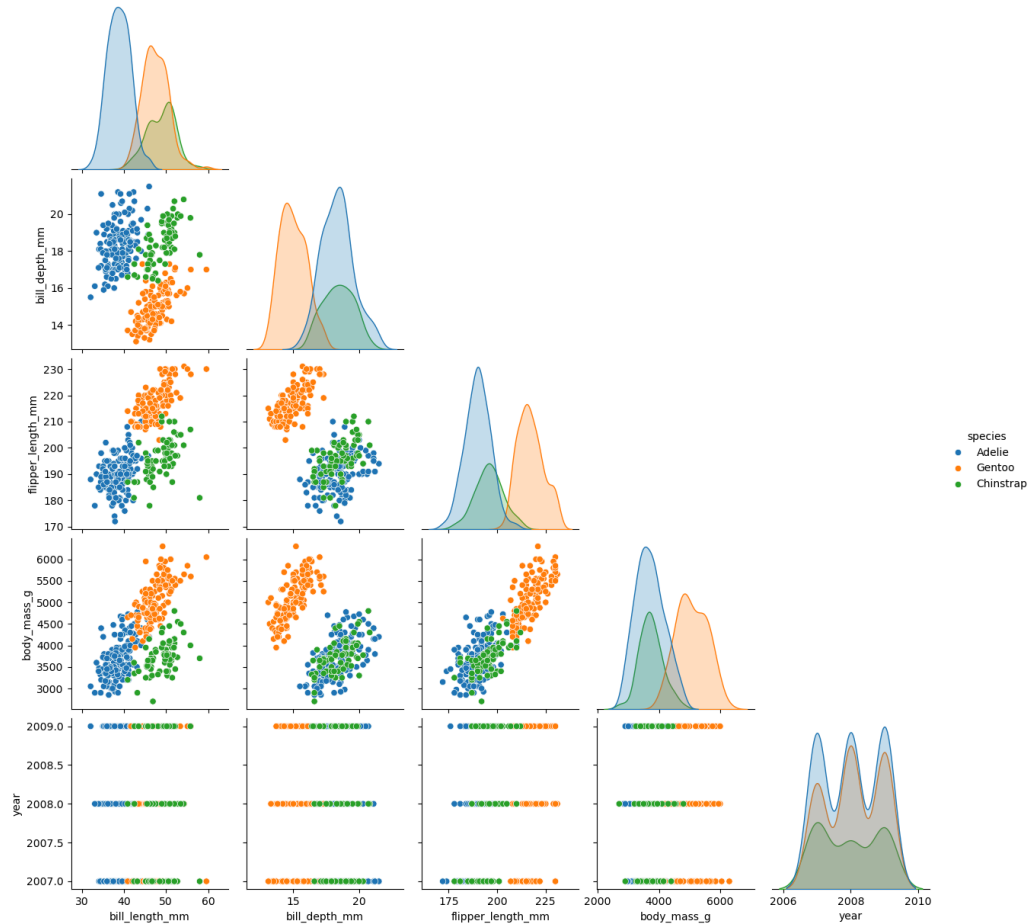
Multivariate analysis

Multivariate analysis explores relationships among three or more variables at once.

Common techniques include:

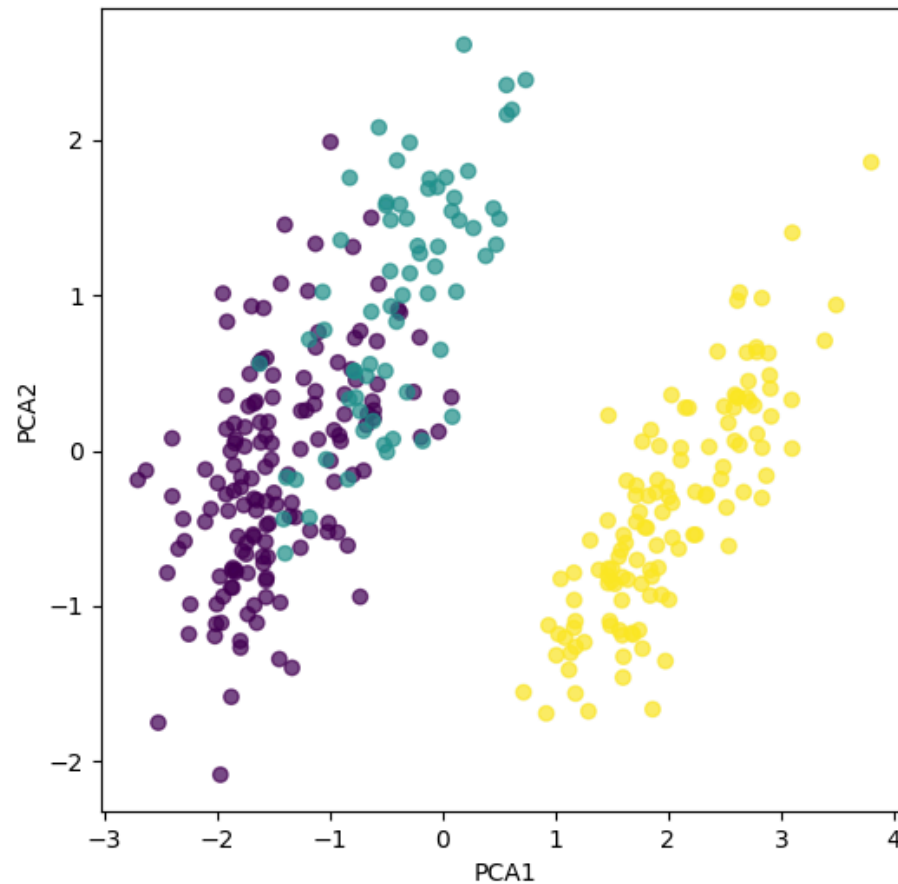
- The same as bivariate analysis, but conditioning on a third variable (e.g., pair plots colored by species).
- Dimensionality reduction methods (e.g., PCA, t-SNE) to visualize high-dimensional data.
- Clustering algorithms (e.g., k-means, hierarchical clustering) to identify groups of similar records.

The last two work on the numerical columns of **X**.



Pair plots of all numerical variables, colored by species.

Colour by species and the clouds line up.



PCA projection of all numerical variables, colored by species.
Two components are enough to separate the species.

From observations to hypotheses

The analysis of the penguins suggests four models:

- Body mass is not a single population, since the sample mixes three species (mixture models, Lecture 4).
- Body mass increases with flipper length (regression, Lecture 6).
- Species and island are dependent, and measurements differ from group to group (hierarchical models, Lecture 4).
- The four measurements are strongly correlated, so fewer dimensions may be enough to describe them (latent variable models, Lecture 4).

The plots establish none of this. They suggest models, which then have to be built, fitted and criticized.

EDA does not stop once a model is built. In the critique step, the same plots come back on what the model gets wrong.

- Residuals reveal patterns the model does not capture.
- Prediction errors point to subsets of the data that are problematic.
- Unexpected patterns suggest new variables or a revised model.

Next lecture: how to draw all of these plots well.

The end.